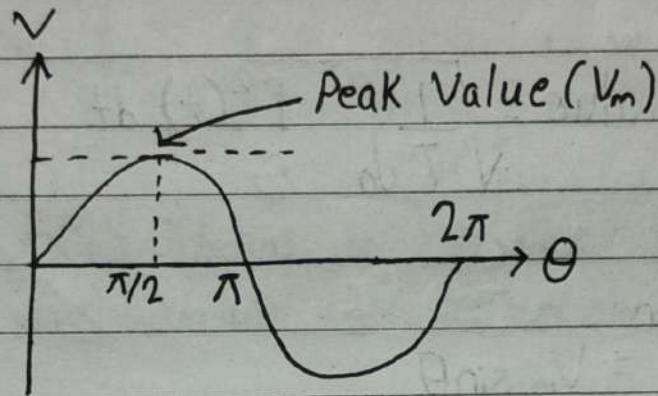


Module - 2: A.C. Circuits

A.C. Fundamentals:

An alternating current supply is that type of supply which changes both in magnitude and direction.



[Here, $\pi = T$]

Q) Find: Average value, RMS value, Peak factor and form factor from above figure.

$$\text{Avg. Value} = \frac{1}{T} \int_0^T f(t) dt \quad [\text{Standard Formula}]$$

For given one,

$$f(t) \sim f(\theta) \quad f(t) = V_m \sin \theta \quad \& \quad T = \pi$$

$$\begin{aligned} \text{So, Avg. Value} &= \frac{1}{\pi} \int_0^{\pi} V_m \sin \theta d\theta \\ &= \frac{V_m}{\pi} \int_0^{\pi} \sin \theta d\theta \\ &= \frac{V_m}{\pi} [-\cos \theta]_0^{\pi} \\ &= \frac{V_m}{\pi} [-\cos \pi + \cos 0] \end{aligned}$$

$$= \frac{2 V_m}{\pi}$$

$$\therefore \boxed{\text{Avg. value} = 0.637 V_m} \quad \text{or} \quad \boxed{V_{\text{avg}} = 0.637 V_m}$$

Similarly, we have formula for RMS value.

$$\text{i.e. RMS value} = \sqrt{\frac{1}{T} \int_0^T f^2(t) dt}$$

For given one,
 $f(\theta) = V_m \sin \theta$

$$\text{So, RMS value} = \sqrt{\frac{1}{\pi} \int_0^{\pi} (V_m \sin \theta)^2 d\theta}$$

$$\text{or, } (RMS)^2 = \frac{V_m^2}{\pi} \int_0^{\pi} \sin^2 \theta d\theta$$

$$= \frac{V_m^2}{\pi} \int_0^{\pi} \left(\frac{1 - \cos 2\theta}{2} \right) d\theta$$

$$= \frac{V_m^2}{2\pi} \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{\pi}$$

$$= \frac{V_m^2}{2\pi} \left\{ \left(\pi - \frac{\sin 2\pi}{2} \right) - \left(0 - \frac{\sin 0}{2} \right) \right\}$$

$$= \frac{V_m^2}{2\pi} (\pi - 0 - 0)$$

$$= \frac{V_m^2}{2\pi} \times \pi$$

$$\text{or, } (RMS)^2 = \frac{V_m^2}{2}$$

$$\therefore V_{RMS}^2 = \frac{V_m^2}{2}$$

$$\therefore \boxed{V_{RMS} = \frac{V_m}{\sqrt{2}}}$$

Form Factor: It is a ratio of RMS value to the Avg. value of alternating quantity.

$$\cancel{K_f} = \frac{\cancel{\text{RMS Value}}}{\cancel{\text{Avg. Value}}} = \frac{\cancel{V_m/\sqrt{2}}}{\cancel{6.37 V_m}} = \frac{\cancel{1}}{\sqrt{2} \times \cancel{6.37}}$$

$$\therefore \cancel{K_f} = 0.11$$

$$K_f = \frac{\text{RMS Value}}{\text{Avg. Value}} = \frac{V_m/\sqrt{2}}{2V_m/\pi} = \frac{\pi}{2\sqrt{2}}$$

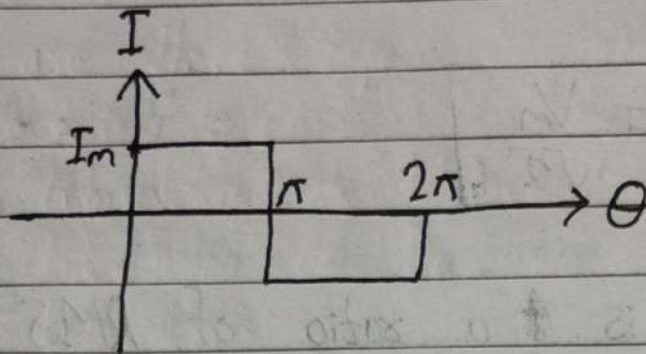
$$\boxed{K_f = 1.11}$$

Peak Factor: It is the ratio of max value of alternating quantity to the RMS value of alternating quantities.

$$K_p = \frac{V_m}{V_m/\sqrt{2}}$$

$$\therefore \boxed{K_p = \sqrt{2}}$$

Q) Find RMS value, Avg. value, Peak factor and Form factor.



$$\begin{aligned}\text{Avg. value} &= \frac{1}{\pi} \int_0^{\pi} I_m d\theta \\ &= \frac{1}{\pi} \times I_m [\theta]_0^{\pi} \\ &= \frac{I_m}{\pi} (\pi - 0)\end{aligned}$$

$$\therefore \text{Avg. value} = I_m$$

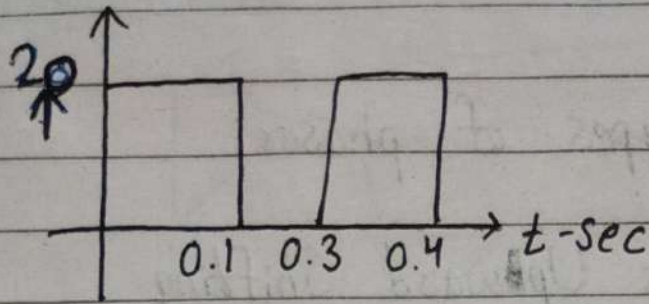
$$\begin{aligned}\text{RMS}^2 &= \frac{1}{\pi} \int_0^{\pi} I_m^2 d\theta \\ &= \frac{I_m^2}{\pi} (\pi - 0)\end{aligned}$$

$$\therefore \text{RMS value} = I_m$$

$$K_p = \frac{I_m}{I_m} = 1$$

$$K_f = \frac{I_m}{I_m} = 1$$

8)



$$T = 0.3$$

$$f(t) = 20$$

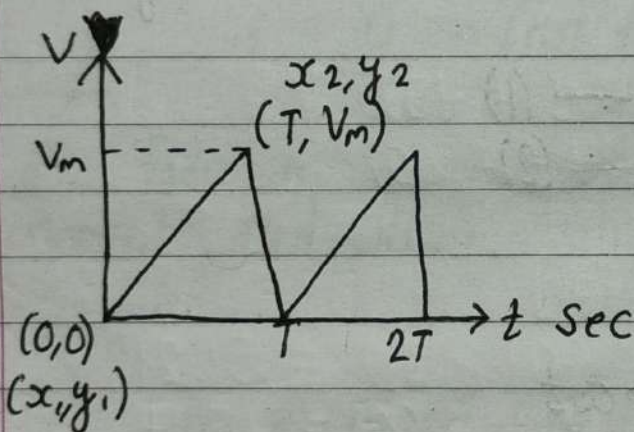
Solⁿ:

$$\text{Avg. value} = \frac{1}{0.3} \int_0^{0.3} 20 \, dt + \text{Cor.}$$

$$= \frac{20}{0.3} [t]_0^{0.3}$$

$$= 6.66 \neq$$

9)



$$T = T$$

$$f(t) = ?$$

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$

$$\frac{V - 0}{T - 0} = \frac{V_m - 0}{T - 0} (t - 0)$$

$$f(t) = V = \frac{V_m t}{T}$$

RMS =

$$\text{Avg. value} = \frac{1}{T} \int_0^{2T} \left(\frac{V_m t}{T} \right)^2 dt$$

$$= \frac{V_m^2}{T^2} \left[\frac{t^3}{3} \right]_0^{2T}$$

$$= \frac{V_m^2}{3T^2} [8T^3 - 0]$$

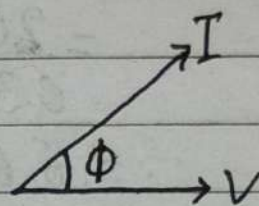
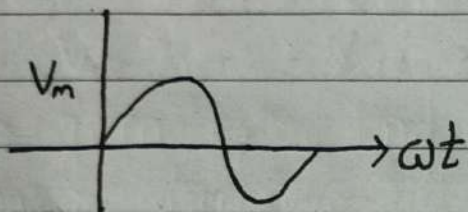
$$= \frac{8V_m^2 T}{3} \neq$$

Types of Phasors

There are three types of phasors:

1. Leading Waveform or Upward Shifting
2. Lagging Waveform or Downward Shifting
3. Unity Waveform

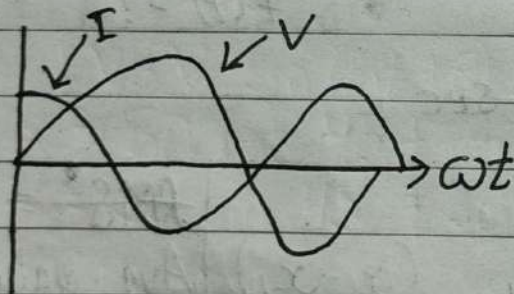
Leading Waveform



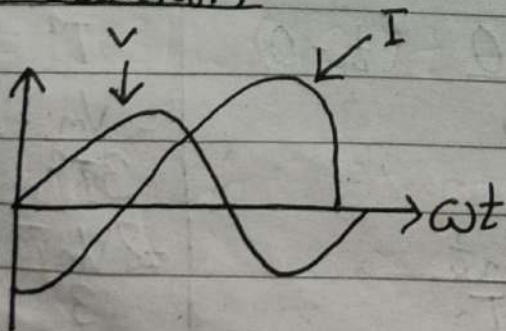
Phasor diagram

$$V = V_m \sin(\omega t + 0) \quad \text{--- (1)}$$

$$I = I_m \sin(\omega t + \phi) \quad \text{--- (2)}$$

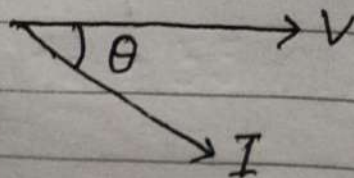


Lagging Waveform

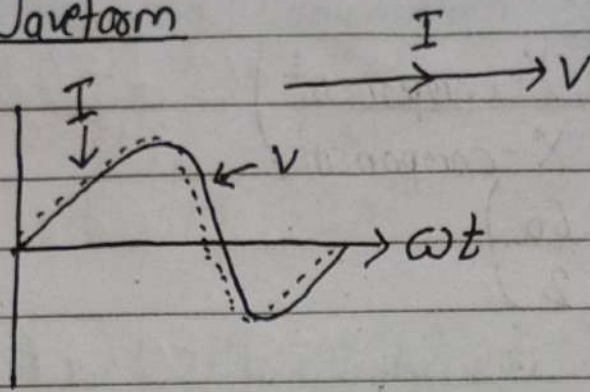


$$V = V_m \sin \omega t \quad \text{--- (1)}$$

$$I = I_m \sin(\omega t - \phi) \quad \text{--- (2)}$$



Unity Waveform



$$V = V_m \sin \omega t \quad \text{--- (1)}$$

$$I = I_m \sin \omega t \quad \text{--- (2)}$$

Q) Two alternating current represented by eq^{ns}.

$$i_1 = 7 \sin \omega t$$

$$i_2 = 10 \sin (\omega t + \pi/3)$$

are fed in a common conductor, find resultant current and RMS value [I=? & RMS=?].

Solⁿ:

$$I_m = \sqrt{(\sum X)^2 + (\sum Y)^2}$$

$\sum X$ = Algebraic sum of 'x' component.

$\sum Y$ = Algebraic sum of 'y' component.

$$\sum X = 7 \cos 0 + 10 \cos (\pi/3) = 12 \text{ A}$$

$$\sum Y = 7 \sin 0 + 10 \sin (\pi/3) = 8.66 \text{ A}$$

$$I_m = \sqrt{(12)^2 + (8.66)^2} = 14.79 \text{ A}$$

$$\therefore I = I_m \sin (\omega t + \phi) = 14.75 \times \sin$$

Now,

$$\phi = \tan^{-1} \left(\frac{Y\text{-component}}{X\text{-component}} \right)$$

$$= \tan^{-1} \left(\frac{8.66}{12} \right)$$

$$= \cancel{0.625}$$

$$= 35.81^\circ$$

$$\therefore I = I_m \sin(\omega t + \phi)$$

$$= 14.75 \times \sin(\omega t + 35.81) \quad \#$$

Q) For given eq^{ns}. find V , frequency, RMS value and Phasor diagram.

$$V_1 = 30 \sin(314t + 45)$$

$$V_2 = 60 \sin(314t + 60)$$

$$\text{Sol}^n: [V = V_m \sin(\omega t + \theta)]$$

$$\Sigma X = 30 \cos 45 + 60 \cos 60 = 51.21$$

$$\Sigma Y = 30 \sin 45 + 60 \sin 60 = 73.17$$

So,

$$V_m = \sqrt{(\Sigma X)^2 + (\Sigma Y)^2} = 89.31 \text{ V}$$

$$\therefore V = 89.31 \sin(314t + \theta)$$

$$\theta = \tan^{-1} \left(\frac{Y\text{-component}}{X\text{-component}} \right) = 55.01^\circ$$

$$\therefore V = 89.31 \sin(314t + 55.01) \quad \underline{\text{Ans}}$$

After comparing any ^{one} given eqⁿ. with $V = V_m \sin(\omega t + \theta)$ we get,

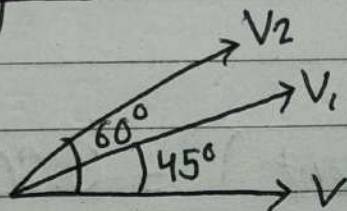
$$314t = \omega t$$

$$\text{or, } 314 = 2\pi f$$

$$\therefore f = \frac{314}{2\pi} = 49.97 \text{ Hz } \underline{\underline{\text{Ans}}}$$

$$\therefore V_{\text{rms}} = \frac{V_m}{\sqrt{2}} = \frac{89.31}{\sqrt{2}} = 63.15 \text{ V}$$

Phasor diagram:



Q) Calculate resultant voltage, frequency, phasor diagram and V_{rms} .

$$V_1 = 100 \sin 500t$$

$$V_2 = 200 \sin (500t + \pi/3)$$

$$V_3 = -50 \cos 500t$$

$$V_4 = 150 \sin (500t - \pi/4)$$

Solⁿ:

$$V_1 = 100 \sin 500t$$

$$V_2 = 200 \sin (500t + \pi/3)$$

$$V_3 = -50 \cos 500t = 50 \sin (500t - \pi/2)$$

$$V_4 = 150 \sin (500t - \pi/4)$$

Now,

$$\begin{aligned}\Sigma X &= 100 \cos 0 + 200 \cos \pi/3 + 50 \cos(-\pi/2) + 150 \cos(-\pi/4) \\ &= 306 \text{ V}\end{aligned}$$

$$\begin{aligned}\Sigma Y &= 100 \sin 0 + 200 \sin \pi/3 + 50 \sin(-\pi/2) + 150 \sin(-\pi/4) \\ &= 17.14 \text{ V}\end{aligned}$$

Then,

$$V_m = \sqrt{(\Sigma X)^2 + (\Sigma Y)^2} = 306.48 \text{ V}$$

$$\therefore V = V_m \sin(500t)$$

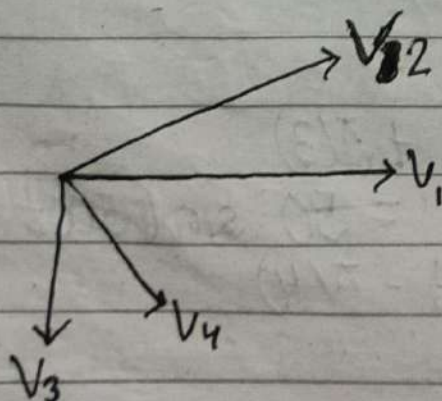
$$\phi = \tan^{-1} \left(\frac{\text{Y-component}}{\text{X-component}} \right) = 3.205^\circ$$

$$\therefore V = 306.48 \sin(500t + 3.2^\circ) \quad \neq$$

Also,

$$\left. \begin{aligned}\omega &= 2\pi f \\ \therefore f &= \frac{\omega}{2\pi}\end{aligned} \right\}$$

$$f = \frac{\omega}{2\pi} = \frac{500}{2\pi} = 79.57 \text{ Hz}$$

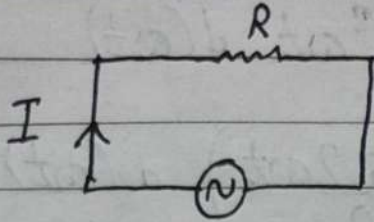


$$V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{306.48}{\sqrt{2}}$$

~~Imp~~

Single Phase A.C. Circuit

① For resistive circuit:



$$V = V_m \sin \omega t \quad \text{--- (1)}$$

We know,

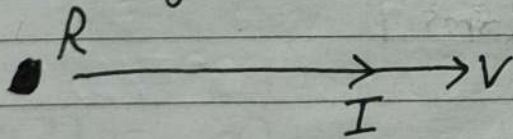
$$I = \frac{V}{R}$$

$$\therefore, I = \frac{V_m \sin \omega t}{R}$$

$$\text{i.e. } I = I_m \sin \omega t \quad \text{--- (2)}$$

From eqⁿ. (1) & (2),

Phasor diagram;



$$\text{Power factor} = \cos \phi = \cos 0 = 1$$

$$\begin{aligned} \text{Avg. Power} &= P_{\text{avg}} = VI \\ &= V_m \sin \omega t \times I_m \sin \omega t \\ &= V_m I_m \sin^2 \omega t \end{aligned}$$

$$\text{i.e. } f(t) = V_m I_m \sin^2 \omega t$$

Now,

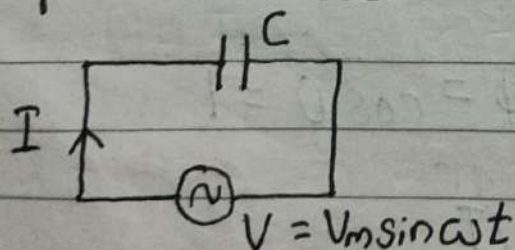
$$\begin{aligned} P_{avg} &= \frac{1}{T} \int_0^T f(t) dt \\ &= \frac{1}{2\pi} \int_0^{2\pi} V_m I_m \sin^2 \omega t d(\omega t) \\ &= \frac{V_m I_m}{2\pi} \int_0^{2\pi} \left(\frac{1 - \cos 2\omega t}{2} \right) d(\omega t) \\ &= \frac{V_m I_m}{2\pi \times 2} \int_0^{2\pi} (1 - \cos 2\omega t) d(\omega t) \\ &= \frac{V_m I_m}{4\pi} \left[\omega t - \frac{\sin 2\omega t}{2} \right]_0^{2\pi} \\ &= \frac{V_m I_m}{4\pi} [2\pi - 0 - 0] \end{aligned}$$

$$P_{avg} = \frac{V_m I_m}{2}$$

$$\text{OR } P_{avg} = \frac{V_m}{\sqrt{2}} \times \frac{I_m}{\sqrt{2}} = V_{rms} I_{rms}$$

$$\therefore P_{avg} = V_{rms} I_{rms}$$

② For capacitive circuit:



$$V = V_m \sin \omega t \quad \text{--- (1)}$$

We know,

$$I = \frac{dq}{dt}$$

$$\text{and, } q = CV = C V_m \sin \omega t$$

$$\text{So, } I = \frac{d}{dt} (C V_m \sin \omega t)$$

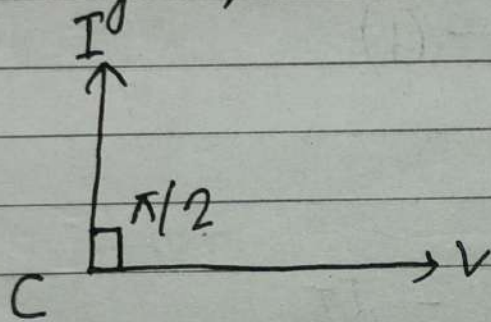
$$\text{or, } I = C V_m \omega \cos \omega t$$

$$\text{or, } I = C V_m \omega \sin(\omega t + \pi/2)$$

$$\text{or, } I = I_m \sin(\omega t + \pi/2) \quad \text{--- (2) [Let, } I_m = C V_m \omega \text{]}$$

From eqn. (1) & (2),

Phasor diagram,



$$\begin{aligned} P &= VI = V_m \sin \omega t \ I_m \sin(\omega t + \pi/2) \\ &= V_m \sin \omega t \times I_m \cos \omega t \\ &= \frac{V_m I_m}{2} 2 \sin \omega t \cos \omega t \end{aligned}$$

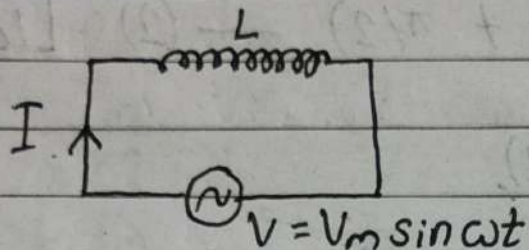
$$f(t) = \frac{V_m I_m}{2} \sin 2\omega t$$

$$\begin{aligned} P_{\text{avg.}} &= \frac{1}{T} \int_0^T f(t) dt \\ &= \frac{1}{2\pi} \int_0^{2\pi} \frac{V_m I_m}{2} \sin 2\omega t \ d(\omega t) \\ &= \frac{V_m I_m}{4\pi} \left[-\frac{\cos 2\omega t}{2} \right]_0^{2\pi} \end{aligned}$$

$$= \frac{V_m I_m}{4\pi} \frac{1}{2} [-1 + 1]$$

$$P_{avg} = 0$$

③ For inductive circuit:



$$\text{let, } V = V_m \sin \omega t \quad \text{--- (1)}$$

We have

$$V = L \frac{dI}{dt}$$

$$\text{or, } \frac{V_m}{L} \sin \omega t \cdot dt = dI$$

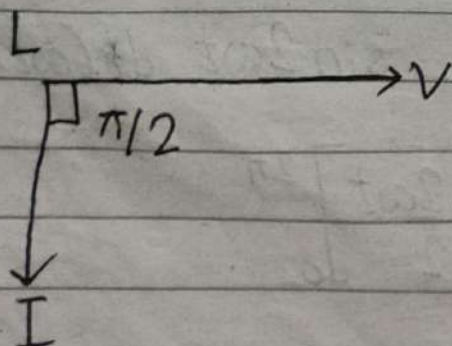
On integrating,

$$I = \frac{V_m}{\omega L} \times -\cos \omega t$$

$$\text{or, } I = + \frac{V_m}{\omega L} \sin (\omega t - \pi/2)$$

$$I = I_m \sin (\omega t - \pi/2) \quad [\text{let } I_m = V_m / \omega L] \quad \text{--- (2)}$$

From eqn. (1) & (2), Phasor diagram,



$$\begin{aligned} \text{Average power} &= VI = V_m \sin \omega t \times I_m \sin(\omega t - \pi/2) \\ (P) &= V_m I_m \sin \omega t \times -\cos \omega t \\ &= -\frac{V_m I_m}{2} 2 \sin \omega t \cos \omega t \end{aligned}$$

$$\sim f(t) = P = -\frac{V_m I_m}{2} \sin 2\omega t$$

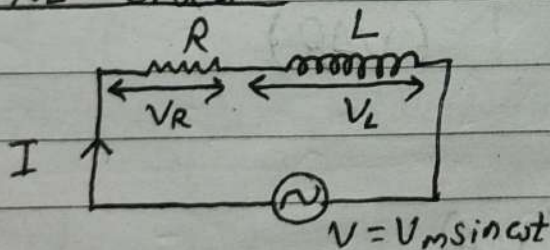
~~where~~

Now,

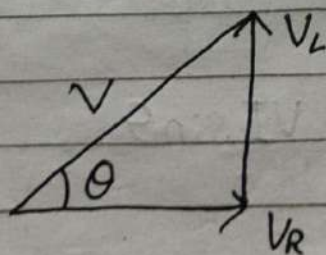
$$\begin{aligned} P_{avg} &= \frac{1}{2\pi} \int_0^{2\pi} -\frac{V_m I_m}{2} \sin 2\omega t \, d(\omega t) \\ &= -\frac{V_m I_m}{4\pi} \left[-\frac{\cos 2\omega t}{2} \right]_0^{2\pi} \\ &= \frac{V_m I_m}{8\pi} (\cos 4\pi - \cos 0) \\ &= \frac{V_m I_m}{8\pi} (1 - 1) \end{aligned}$$

$$P_{avg} = 0$$

④ For RL circuit:



Phasor diagram



From figure,

$$V = \sqrt{V_R^2 + V_L^2}$$

$$= \sqrt{(IR)^2 + (IX_L)^2}$$

$$V = I \sqrt{R^2 + X_L^2}$$

$$\therefore Z = \frac{V}{I} = \sqrt{R^2 + X_L^2}$$

where, Z is impedance.

$$\theta = \tan^{-1} \left(\frac{V_L}{V_R} \right) = \tan^{-1} \left(\frac{IX_L}{IR} \right)$$

$$\therefore \theta = \tan^{-1} \left(\frac{X_L}{R} \right)$$

Types of Power

1. Active/Real/True Power:

$$P = VI \cos \theta \quad (\text{Watt})$$

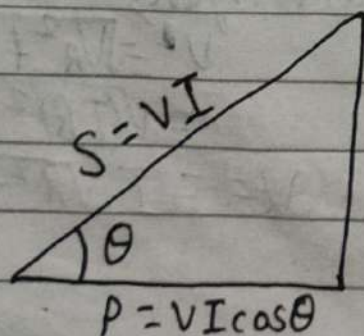
2. Reactive Power:

$$Q = VI \sin \theta \quad (\text{VAR})$$

3. Apparent Power:

$$S = VI \quad (\text{VA})$$

Power Triangle:



$$Q = VI \sin \theta$$

$$S^2 = P^2 + Q^2$$

Power Factor

The cosine angle between voltage and current is called power factor.

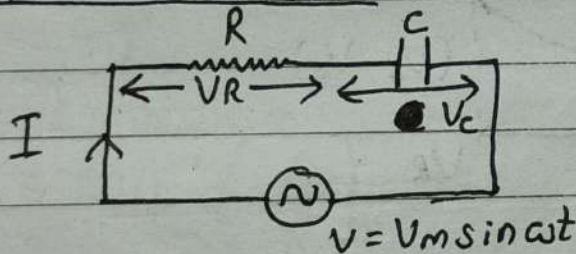
→ Using Impedance:

$$\cos \phi = \frac{R}{Z}$$

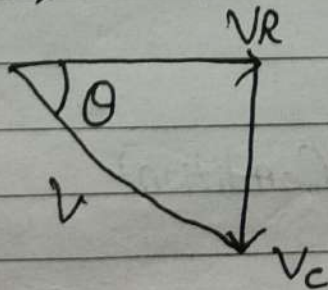
→ Using Power Triangle:

$$\text{Power Factor} = \frac{\text{Real Power}}{\text{Apparent Power}}$$

⑤ For R-C circuit:



Phasor diagram,



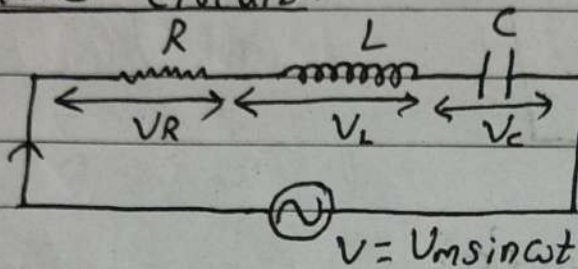
$$V = \sqrt{V_R^2 + V_C^2}$$

$$Z = \frac{V}{I} = \sqrt{R^2 + X_C^2}$$

$$\theta = \tan^{-1} \left(\frac{V_C}{V_R} \right)$$

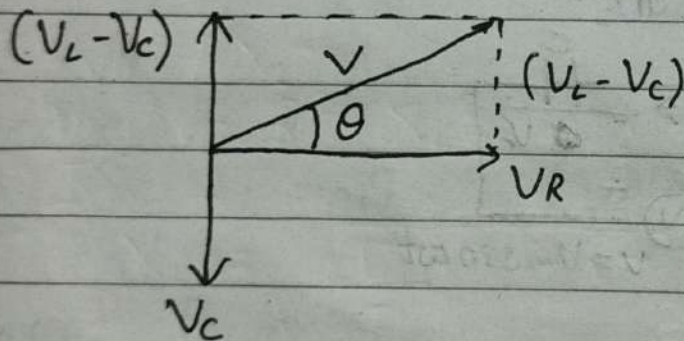
$$\therefore \theta = \tan^{-1} \left(\frac{X_C}{X_R} \right)$$

⑥ For RLC circuit:



~~Phasor diagram~~

* (i) $V_L > V_C \Rightarrow V_L - V_C$



(ii) $V_C > V_L \Rightarrow V_C - V_L$

(iii) $V_L = V_C$ (Resonance Condition)

For (i),

$$V = \sqrt{(V_L - V_C)^2 + V_R^2}$$

$$Z = \frac{V}{I} = \sqrt{(X_L - X_C)^2 + R^2}$$

$$\theta = \tan^{-1} \left(\frac{X_L - X_C}{R} \right)$$

For (ii),

$$Z = \frac{V}{I} = \sqrt{(X_C - X_L)^2 + R^2}$$

$$\theta = \tan^{-1} \left(\frac{X_C - X_L}{R} \right)$$

For (iii),

$$Z = \frac{V}{I} = R$$

$$\theta = \tan^{-1} \left(\frac{0}{R} \right) = 0$$

Q) An alternating current with current 1.5 A , voltage 200 V . The consumed power is 225 W . Find resistance and inductive reactance of the circuit.

Solⁿ:

Given, $I = 1.5\text{ A}$
 $V = 200\text{ V}$
 $P = 225\text{ W}$
 $R = ?$
 $X_L = ?$

We have

$$P = I^2 R$$
$$\therefore 225 = (1.5)^2 \times R$$
$$\therefore R = \frac{225}{(1.5)^2} = 100\ \Omega$$

$$Z = \sqrt{R^2 + X_L^2}$$
$$\therefore \left(\frac{V}{I}\right)^2 = R^2 + X_L^2$$
$$\therefore X_L = \sqrt{\left(\frac{V}{I}\right)^2 - R^2}$$
$$= 88.19\ \Omega$$

Q) A coil takes a current of 2.5 A , when connected across 250 V , 50 Hz and consumed power 400 W . Calculate power factor, resistance of the coil and inductance.

Solⁿ:

$$I = 2.5\text{ A}$$

$$V = 250\text{ V}$$

$$f = 50\text{ Hz}$$

$$P = 400\text{ W}$$

$$\cos \phi = ?$$

$$R = ?$$

$$L = ?$$

$$P = I^2 R$$

$$\therefore R = \frac{P}{I^2} = \frac{400}{(2.5)^2} = 64\ \Omega //$$

$$Z = \frac{V}{I} = \frac{250}{2.5} = 100\ \Omega$$

$$\therefore \cos \phi = \frac{R}{Z} = \frac{64}{100} = 0.64 //$$

$$Z^2 = R^2 + X_L^2$$

$$\Rightarrow X_L = \sqrt{Z^2 - R^2} = 76.83\ \Omega$$

$$\therefore L = \frac{X_L}{2\pi f} = 0.24\text{ H} // \quad [X_L = \omega L]$$

Q) A coil having a resistance of $10\ \Omega$ and inductance of 0.5 H is connected in series with a capacitor of $100\ \mu\text{F}$. The circuit has been connected to a 200 V , 50 Hz single phase AC supply. Calculate impedance, current, power factor, active power, reactive power and apparent power. Also find voltage across resistor, inductor and capacitor.

Solⁿ:

$$R = 10\ \Omega$$

$$L = 0.5\text{ H}$$

$$C = 100\ \mu\text{F} = 100 \times 10^{-6}\text{ F}$$

$$V = 200\text{ V}$$

$$f = 50\text{ Hz}$$

$$Z = ?$$

$$I = ?$$

$$\cos \phi = ?$$

$$\text{Active power} = ?$$

$$\text{Reactive power} = ?$$

$$\text{Apparent power} = ?$$

$$V_R = ?$$

$$V_L = ?$$

$$V_C = ?$$

$$X_L = \omega L = 2\pi f L = 2 \times \pi \times 50 \times 0.5 = 157.079\ \Omega$$

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C} = \frac{1}{2 \times \pi \times 50 \times 100 \times 10^{-6}} = 31.83\ \Omega$$

$$\begin{aligned}\therefore Z &= \sqrt{(X_L - X_C)^2 + R^2} \\ &= \sqrt{(157.079 - 31.83)^2 + 10^2} \\ &= 125.64 \, \Omega\end{aligned}$$

$$Z = \frac{V}{I}$$

$$\therefore I = \frac{V}{Z} = \frac{200}{125.64} = 1.5918 \, A$$

$$\therefore \cos \phi = \frac{R}{Z} = \frac{10}{125.64} = 0.0795$$

$$\begin{aligned}\therefore V_R &= IR = 1.59 \times 10 = 15.9 \, V \\ V_L &= IX_L = 1.59 \times 157.07 = 249.74 \, V \\ V_C &= IX_C = 1.59 \times 31.83 = 50.60 \, V\end{aligned}$$

$$\begin{aligned}\text{Active Power} &= VI \cos \phi \\ &= 200 \times 1.5918 \times 0.0795 \\ &= 25.309 \, W\end{aligned}$$

$$\begin{aligned}\text{Reactive Power} &= VI \sin \phi \\ &= VI \sqrt{1 - \cos^2 \phi} \\ &= 200 \times 1.5918 \times \sqrt{1 - (0.0795)^2} \\ &= 317.35 \, W\end{aligned}$$

$$\begin{aligned}\text{Apparent Power} &= VI \\ &= 200 \times 1.5918 \\ &= 318.36 \, W\end{aligned}$$

Imp # Resonance

There will be value of frequency at which the current is maximum, exist in the electrical resonance (series resonance).

RLC series resonance frequency: in case

$$V_L = V_C \quad (\text{At resonance})$$

$$\text{or, } I_m X_L = I_m X_C$$

$$\text{or, } X_L = X_C$$

$$\text{or, } \omega_r L = \frac{1}{\omega_r C}$$

$$\text{or, } \omega_r^2 = \frac{1}{LC}$$

$$\text{or, } \omega_r = \frac{1}{\sqrt{LC}}$$

$$\text{or, } 2\pi f_r = \frac{1}{\sqrt{LC}}$$

$$\therefore \boxed{f_r = \frac{1}{2\pi\sqrt{LC}}}$$

Relation Between Upper-Cutoff Frequency & Lower-Cutoff Frequency

$$\omega_r = \sqrt{\omega_1 \cdot \omega_2} \quad \text{or} \quad f_r = \sqrt{f_1 \cdot f_2}$$

$$\text{Bandwidth} = f_2 - f_1 = \frac{R}{2\pi L}$$

Let,

$$I_{rms} = \frac{V}{\sqrt{R^2 + (X_L - X_C)^2}} \quad \text{--- (1)}$$

At series resonance condition, current is maximum i.e. I_m .

$$I_m = \frac{V}{R}$$

$$\text{or, } \sqrt{2} I_{rms} = \frac{V}{R} \quad \left[\because I_{rms} = \frac{I_m}{\sqrt{2}} \right]$$

$$\text{or, } I_{rms} = \frac{V}{\sqrt{2} R} \quad \text{--- (2)}$$

From eqⁿ. (1) & (2),

$$\frac{V}{\sqrt{R^2 + (X_L - X_C)^2}} = \frac{V}{R\sqrt{2}}$$

$$\text{or, } \sqrt{R^2 + (X_L - X_C)^2} = R\sqrt{2}$$

Squaring both sides,

$$R^2 + (X_L - X_C)^2 = 2R^2$$

$$\text{or, } (X_L - X_C)^2 = R^2$$

$$\text{or, } X_L - X_C = \pm R$$

$$\text{or, } \omega L - \frac{1}{\omega C} = \pm R$$

$$\omega_2 L - \frac{1}{\omega_2 C} = +R \quad \text{--- (3)}$$

$$\omega_1 L - \frac{1}{\omega_1 C} = -R \quad \text{--- (4)}$$

Adding eqns. (3) & (4),

$$(\omega_1 + \omega_2)L - \frac{1}{C} \left(\frac{1}{\omega_1} + \frac{1}{\omega_2} \right) = 0$$

$$\text{or, } L(\omega_1 + \omega_2) - \frac{1}{C} \left(\frac{\omega_2 + \omega_1}{\omega_1 \omega_2} \right) = 0$$

$$\text{or, } (\omega_1 + \omega_2)L = \frac{1}{C} \left(\frac{\omega_1 + \omega_2}{\omega_1 \omega_2} \right)$$

$$\text{or, } \omega_1 \times \omega_2 = \frac{1}{LC} \quad \text{--- (5)}$$

In series resonance,

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$\text{or, } 2\pi f_r = \frac{1}{\sqrt{LC}}$$

$$\text{or, } \omega_r^2 = \frac{1}{LC} \quad \text{--- (6)}$$

From (5) & (6),

$$\omega_s^2 = \omega_1 \omega_2$$

$$\therefore \boxed{\omega_s = \sqrt{\omega_1 \omega_2}}$$

$$\boxed{f_s = \sqrt{f_1 \cdot f_2}}$$

Exⁿ. (3) - (4),

$$(\omega_2 - \omega_1)L - \frac{1}{C} \left(\frac{1}{\omega_2} - \frac{1}{\omega_1} \right) = R(-R) = 2R \quad \text{--- (7)}$$

Dividing both sides by inductor,

$$(\omega_2 - \omega_1) - \frac{1}{LC} \left(\frac{\omega_1 - \omega_2}{\omega_1 \omega_2} \right) = \frac{2R}{L}$$

From eqⁿ. (5), $\frac{1}{LC} = \omega_1 \omega_2$

$$(\omega_2 - \omega_1) - \omega_1 \omega_2 \left(\frac{\omega_1 - \omega_2}{\omega_1 \omega_2} \right) = \frac{2R}{L}$$

$$\text{or, } \omega_2 - \omega_1 - \omega_1 + \omega_2 = 2R/L$$

$$\text{or, } 2(\omega_2 - \omega_1) = 2R/L$$

$$\text{or, } \omega_2 - \omega_1 = R/L$$

$$\text{or, } 2\pi f_2 - 2\pi f_1 = R/L$$

$$\text{or, } 2\pi(f_2 - f_1) = R/L$$

$$\therefore \boxed{f_2 - f_1 = \frac{R}{2\pi L}}$$

Imp # Quality Factor (Q_s)

$Q_s = \frac{\text{Voltage across Inductance or Capacitance}}{\text{Supply Voltage}}$

Quality factor is the ratio of voltage across inductance or capacitance and supply voltage.

$$\text{i.e. } Q_s = \frac{V_L \text{ or } V_C}{V}$$

$$(i) \quad \cancel{E_{os}} \quad Q_s = \frac{V_L}{V} = \frac{I_m X_L}{I_m R} = \frac{\omega_s L}{R} = \frac{2\pi f_s L}{R}$$

$$\therefore \boxed{Q_s = \frac{1}{R} \sqrt{\frac{L}{C}}}$$

$$(ii) \quad Q_s = \frac{V_C}{V} = \frac{I_m X_C}{I_m R} = \frac{X_C}{R} = \frac{1}{\omega_s RC}$$

$$\therefore \boxed{Q_s = \frac{1}{R} \sqrt{\frac{L}{C}}}$$

Rectangular Form To Polar Form

$$Z = 5 + j5$$

$$|Z| = \sqrt{5^2 + 5^2} = \sqrt{50} = 7.07$$

$$\theta = \tan^{-1} \left(\frac{I_p}{R_p} \right) = \tan^{-1} \left(\frac{5}{5} \right) = 45$$

$$\therefore \text{Polar form} = 7.07 \angle 45$$

Q) Solve: $(5 + j5)(10 \angle 45)$

Solⁿ: $(5 + j5)(10 \angle 45)$

$$\begin{aligned} &= (7.07 \angle 45)(10 \angle 45) \\ &= (70.7 \angle 90) \end{aligned}$$

Q) Solve: $\frac{(7.07 \angle 45)}{(10 \angle 45)}$

Solⁿ: $\frac{(7.07 \angle 45)}{(10 \angle 45)}$
 $= (0.707 \angle 0)$

Q) Find $\frac{AB}{C}$ in Rectangular form.

$$A = 8 + j6$$

$$B = 30 \angle 55$$

$$C = 10 + j7$$

Solⁿ:

$$A = 8 + j6 = 10 \angle 36.869$$

$$B = 30 \angle 55$$

$$C = 10 + j7 = 12.2 \angle 34.992$$

Now,

$$\frac{AB}{C} = \frac{(10 \angle 36.869)(30 \angle 55)}{(12.2 \angle 34.992)}$$

$$= \frac{(300 \angle 91.869)}{(12.2 \angle 34.992)}$$

$$= \frac{24.59 \angle 56.877}{1}$$

$$= 24.59 \angle 56.877$$

$$= 13 + j21 \quad \#$$

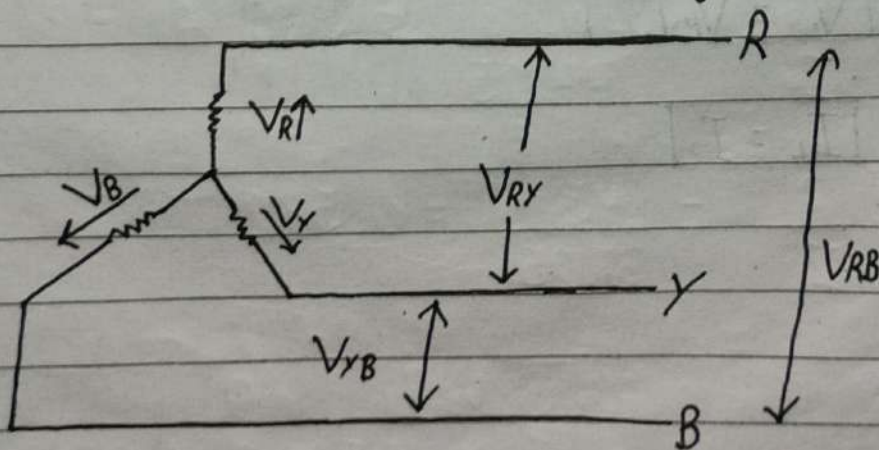
3-Phase Balanced AC System

Advantages of 3- ϕ balanced ac system over single-phase:

1. Three phase contains the rotating magnetic field but in single phase doesn't contain the rotating magnetic field.
2. 3-phase has minimum conductor material but in single phase maximum conductor material.
3. Efficiency of 3-phase machine is greater than single-phase machine.
4. 3-phase supply is used in generation, transmission and distribution & also utilization of electrical energy but in single phase only utilization of electrical energy.

Imp. #

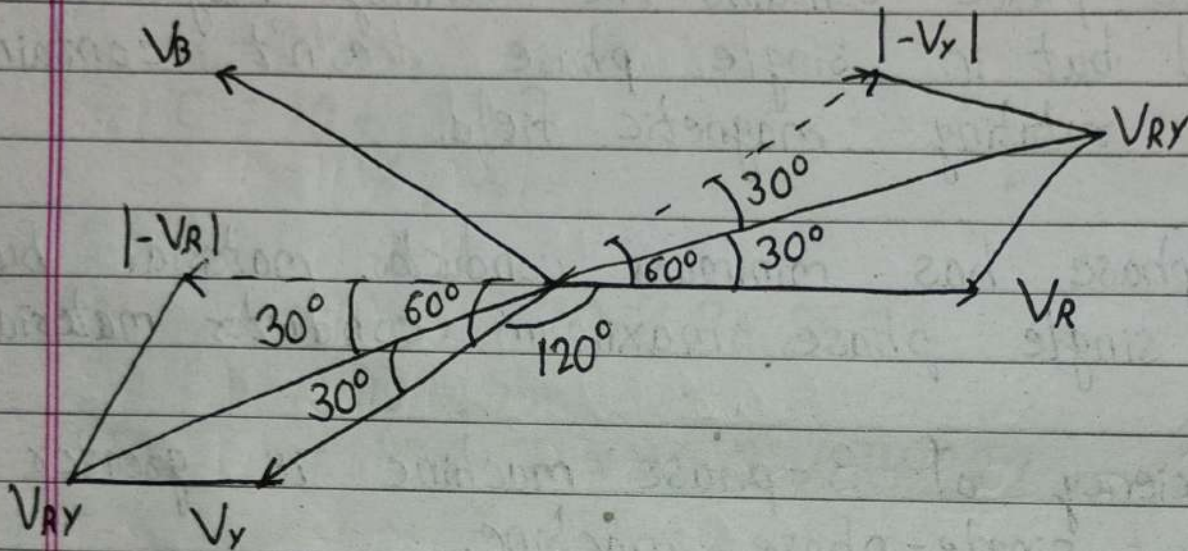
3-Phase (ϕ) Balanced AC System - Star Connected



$$V_R = V_Y = V_B = V_P \text{ (Phase Voltage)}$$

$$V_{RY} = V_{YB} = V_{RB} = V_L \text{ (Line Voltage)}$$

Phasor Diagram,



From $\triangle V_{RY}$,

$$V_{RY}^2 = V_R^2 + V_Y^2 + 2V_R V_Y \cos 60^\circ$$

$$\text{or, } V_L^2 = V_P^2 + V_P^2 + 2V_P V_P \cdot \frac{1}{2}$$

$$\text{or, } V_L^2 = 3V_P^2$$

$$\therefore \boxed{V_L = V_P \sqrt{3}}$$

$$\Rightarrow \boxed{I_L = I_P}$$

Types of Power (For 3-phase)

1. Active Power,

$$P = 3 V_p I_p \cos \theta \quad (\text{Watt})$$

$$\text{or, } P = 3 \frac{V_L}{\sqrt{3}} I_L \cos \theta$$

$$\text{or, } P = \sqrt{3} \sqrt{3} \frac{V_L}{\sqrt{3}} I_L \cos \theta$$

$$\therefore P = \sqrt{3} V_L I_L \cos \theta$$

2. Reactive Power,

$$P = 3 V_p I_p \sin \theta \quad (\text{VAR})$$

$$P = \sqrt{3} V_L I_L \sin \theta$$

3. Apparent Power,

$$S = 3 V_p I_p \quad (\text{VA})$$

$$S = \sqrt{3} V_L I_L$$

Q) Three coils with resistance 45Ω and inductance of 0.2 H are connected with 415 V supply. Three-Phase supply at 50 Hz , star connected. Calculate the current in each coil and active power, reactive power and apparent power.

Solⁿ:

$$R = 45 \Omega$$

$$L = 0.2 \text{ H}$$

$$V_L = 415 \text{ V}$$

$$f = 50 \text{ Hz}$$

$$V_p = \frac{V_L}{\sqrt{3}} = \frac{415}{\sqrt{3}} = 239.6 \text{ V}$$

$$Z_p = \frac{V_p}{I_p}$$

$$\therefore I_p = \frac{V_p}{Z_p}$$

$$X_L = 2\pi fL = 62.8 \Omega$$

$$Z_p = \sqrt{R^2 + X_L^2} = \sqrt{45^2 + (62.8)^2} = 77.25 \Omega$$

$$\therefore I_p = \frac{V_p}{Z_p} = \frac{239.6}{77.25} = 3.10 \text{ A}$$

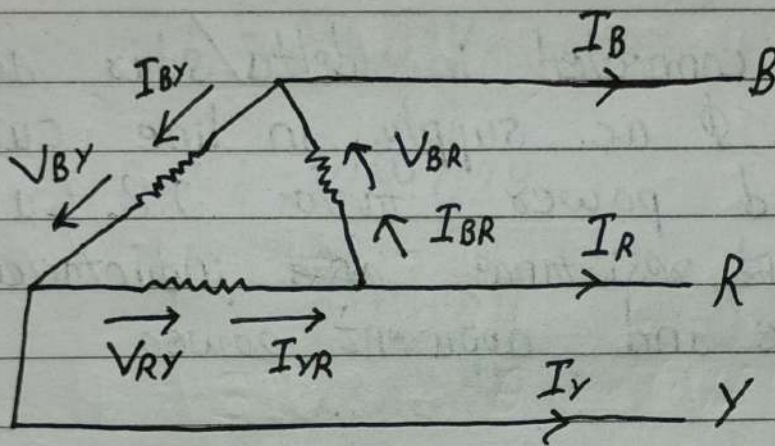
$$\cos \phi = \frac{R}{Z_p} = \frac{45}{77.25} = 0.582$$

$$\begin{aligned} \therefore \text{Reactive Power} &= \sqrt{3} V_L I_p \sin \theta \\ &= \sqrt{3} \times 415 \times 3.10 \times 0.8131 \\ &= 1811.81 \text{ W} \end{aligned}$$

$$\begin{aligned}
 \text{Active Power } (P) &= \sqrt{3} V_L I_P \cos \phi \\
 &= \sqrt{3} \times 415 \times 3.10 \times 0.582 \\
 &= 1296.86 \text{ VAR}
 \end{aligned}$$

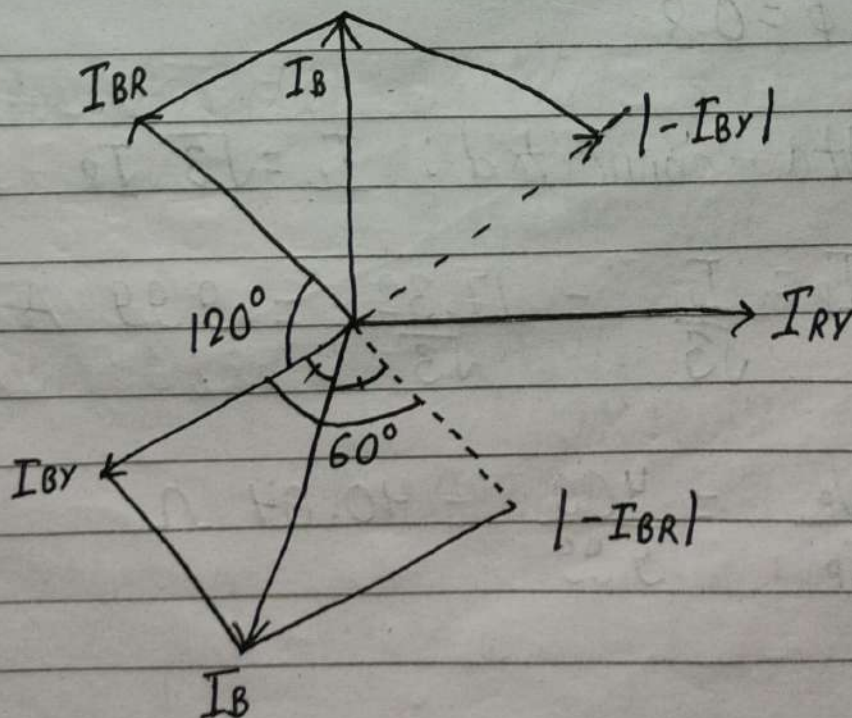
$$\begin{aligned}
 \text{Apparent Power } (P) &= \sqrt{3} V_L I_P \\
 &= \sqrt{3} \times 415 \times 3.10 \\
 &= 2228.28 \text{ VA}
 \end{aligned}$$

3 ϕ Balanced (Delta Connected) AC System



$$I_{BR} = I_{BY} = I_{RY} = I_P \text{ (Phase Current)}$$

$$I_B = I_R = I_Y = I_L \text{ (Line Current)}$$



From 11^gm

$$I_B^2 = I_{BR}^2 + I_{BY}^2 + 2 I_{BR} I_{BY} \cos 60^\circ$$
$$\therefore, I_L^2 = I_P^2 + I_P^2 + 2 \cdot I_P \cdot I_P \cdot \frac{1}{2}$$

$$\therefore, I_L^2 = 3 I_P^2$$

$$\therefore \boxed{I_L = \sqrt{3} I_P}$$

$$\Rightarrow \boxed{V_L = V_P}$$

- Q) Three coils connected in delta/star across 400 V, 50 Hz. Three ϕ ac. supply in line current of 17.32 A and power factor 0.8. Calculate phase current, resistance, ~~and~~ inductance and active, reactive and apparent power.

Solⁿ:

$$V_L = 400 \text{ V}$$

$$f = 50 \text{ Hz}$$

$$I_L = 17.32 \text{ A}$$

$$\cos \phi = 0.8$$

- ① For delta-connected: $I_L = \sqrt{3} I_P$ & $V_L = V_P$

$$I_P = \frac{I_L}{\sqrt{3}} = \frac{17.32}{\sqrt{3}} = 9.99 \text{ A}$$

$$Z_P = \frac{V_P}{I_P} = \frac{400}{9.99} = 40.04 \Omega$$

$$\cos \phi = \frac{R}{Z_P}$$

$$\therefore R = 0.8 \times 40.04 = 32.032 \, \Omega$$

$$Z_P = \sqrt{R^2 + X_L^2}$$

$$\text{or, } \sqrt{Z_P^2 - R^2} = X_L$$

$$\text{or, } X_L = \sqrt{(40.04)^2 - (32.032)^2} = 24.024$$

$$\text{or, } \omega L = 24.024$$

$$\text{or, } 2\pi f L = 24.024$$

$$\therefore L = \frac{24.024}{2\pi \times 50} = 0.076509 \, \text{H}$$

$$\begin{aligned} \text{Active power (P)} &= \sqrt{3} V_L I_L \cos \phi \\ &= \sqrt{3} \times 400 \times 17.32 \times 0.8 \\ &= 9,599.71 \, \text{Watt} \end{aligned}$$

$$\begin{aligned} \text{Reactive Power (P)} &= \sqrt{3} V_P I_P \sin \phi \\ &= \sqrt{3} \times 400 \times 9.99 \times 0.6 \\ &= 4,152.76 \, \text{VAR} = 7,192.8 \, \text{VAR} \end{aligned}$$

$$\begin{aligned} \text{Apparent Power} &= \sqrt{3} \times V_L I_L \\ &= \sqrt{3} \times 400 \times 17.32 \\ &= 11,999.64 \, \text{VA} \end{aligned}$$

② For star-connected: $V_L = V_p \sqrt{3}$ & $I_L = I_p$

$$V_p = \frac{V_L}{\sqrt{3}} = \frac{400}{\sqrt{3}} = 230.94 \text{ V}$$

$$Z_p = \frac{V_p}{I_p} = \frac{230.94}{17.32} = 13.33 \text{ V}$$

$$\cos \phi = \frac{R}{Z_p}$$

$$\therefore R = 0.8 \times 13.33 = 10.664 \Omega$$

$$Z_p = \sqrt{R^2 + X_L^2} =$$

$$\therefore X_L = \sqrt{Z_p^2 - R^2} = \sqrt{(13.33)^2 - (10.664)^2} = 8.0019$$

$$\therefore 2\pi f L = 8.0019$$

$$\therefore L = \frac{8.0019}{2\pi \times 50} = 0.0254 \text{ H}$$

$$\text{Active Power} = \sqrt{3} V_L I_L \cos \phi = \sqrt{3} \times 400 \times 17.32 \times 0.8 = 9,599.718 \text{ W}$$

$$\text{Reactive Power} = \sqrt{3} V_p I_p \sin \phi = \sqrt{3} \times 230.94 \times 17.32 \times 0.6 = 4,156.798 \text{ VAR}$$

$$\text{Apparent Power} = \sqrt{3} V_L I_L = \sqrt{3} \times 400 \times 17.32 = 11,999.647 \text{ VA}$$

Power measurement in 3ϕ AC circuit

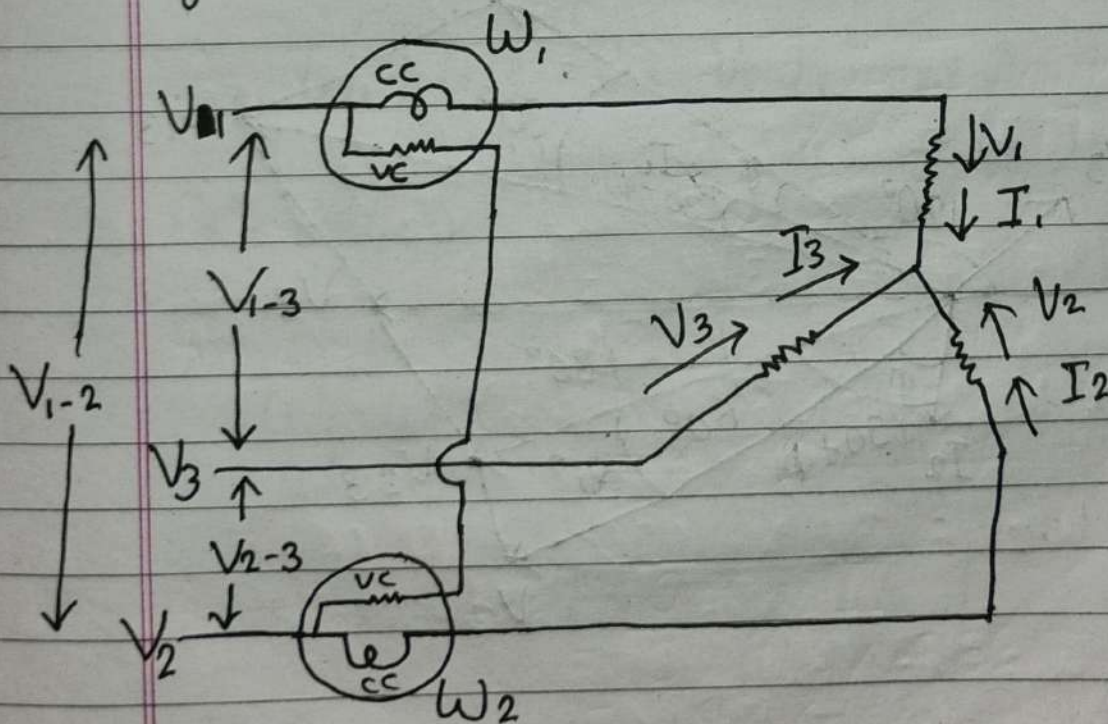
Power measurement in 3ϕ balanced a.c. circuit by the method of Watt meter.

Watt meter is a combination of an ^{am-}meter and voltmeter.

There are three types watt meter:

- ① 1-Wattmeter
- ✓ ② 2-Wattmeter
- ③ 3-Wattmeter

- Power measurement by 2-watt meter when system is star connected.



$$P = V_1 I_1 + V_2 I_2 + V_3 I_3$$

At Balanced Condition,

$$I_1 + I_2 + I_3 = 0$$

$$\Rightarrow I_3 = -I_1 - I_2$$

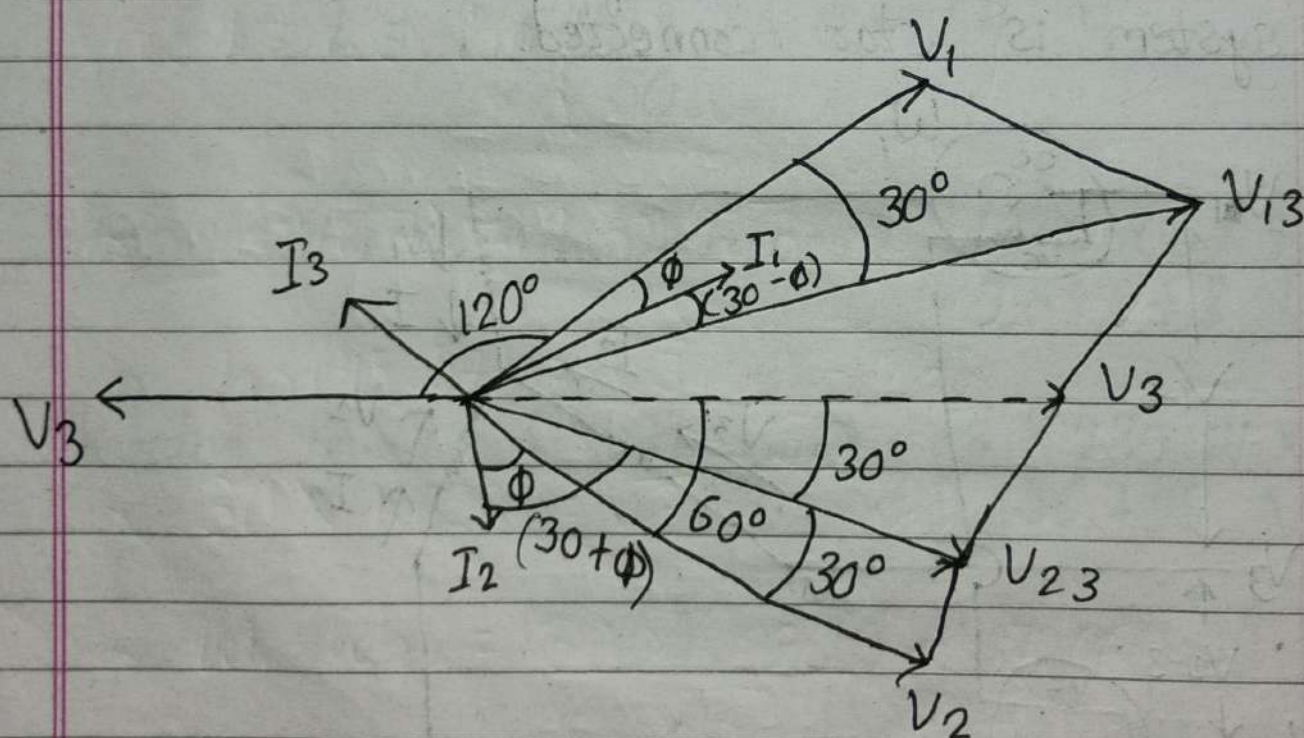
$$P = V_1 I_1 + V_2 I_2 + V_3 (-I_1 - I_2)$$

$$\text{or, } P = V_1 I_1 + V_2 I_2 - V_3 I_1 - I_2 V_3$$

$$\text{or, } P = I_1 (V_1 - V_3) + I_2 (V_2 - V_3)$$

$$\therefore \boxed{P = W_1 + W_2}$$

- Determine Power Factor ^{using} two watt-meter method.



For W_1 ,

$$W_1 = V_{13} I_1 \cos(30^\circ - \phi)$$

$$\text{or, } W_1 = V_L I_L \cos(30^\circ - \phi) \quad \text{--- (1)}$$

For W_2 ,

$$W_2 = V_{L3} I_2 \cos(30^\circ + \phi)$$

$$\text{or, } W_2 = V_L I_L \cos(30^\circ + \phi) \quad \text{--- (2)}$$

Eqⁿ. (1) - (2),

$$\begin{aligned} W_1 - W_2 &= V_L I_L [\cos(30^\circ - \phi) - \cos(30^\circ + \phi)] \\ &= V_L I_L [2 \sin 30^\circ \cdot \sin \phi] \\ &= V_L I_L \cdot \frac{2 \times 1}{2} \sin \phi \end{aligned}$$

$$\text{or, } W_1 - W_2 = V_L I_L \sin \phi \quad \text{--- (3)}$$

Eqⁿ. (1) + (2),

$$\begin{aligned} W_1 + W_2 &= V_L I_L [\cos(30^\circ - \phi) + \cos(30^\circ + \phi)] \\ &= V_L I_L [2 \cos 30^\circ \cos \phi] \\ &= V_L I_L \cdot 2 \cdot \frac{\sqrt{3}}{2} \cos \phi \end{aligned}$$

$$\text{or, } W_1 + W_2 = \sqrt{3} V_L I_L \cos \phi \quad \text{--- (4)}$$

Eqⁿ. $\frac{(3)}{(4)}$,

$$\frac{W_1 - W_2}{W_1 + W_2} = \frac{V_L I_L \sin \phi}{\sqrt{3} V_L I_L \cos \phi}$$

$$\text{or, } \tan \phi = \sqrt{3} \left[\frac{W_1 - W_2}{W_1 + W_2} \right]$$

$$\text{or, } \phi = \tan^{-1} \left[\sqrt{3} \left(\frac{W_1 - W_2}{W_1 + W_2} \right) \right]$$

$$\therefore \boxed{\text{Power factor} = \cos \phi = \cos \tan^{-1} \left[\frac{\sqrt{3} (W_1 - W_2)}{W_1 + W_2} \right]}$$

Q) In a two wattmeter method, the total power is 30 kW and 0.7 power factor. Calculate reading of each wattmeter.

Solⁿ:

$$P = 30 \text{ kW}$$

$$\cos \phi = 0.7$$

We know,

$$P = W_1 + W_2$$

$$\Rightarrow W_2 = 30 - W_1$$

$$\cos \phi = \cos \tan^{-1} \left(\frac{\sqrt{3} (W_1 - W_2)}{W_1 + W_2} \right)$$

$$\therefore, \cos^{-1}(0.7) = \tan^{-1} \left(\frac{\sqrt{3} (W_1 - W_2)}{W_1 + W_2} \right)$$

$$\therefore, \tan(45.57) = \frac{\sqrt{3} (W_1 - W_2)}{30}$$

$$\therefore, 30 \times 1.020 = \sqrt{3} (W_1 - W_2)$$

$$\therefore, W_1 - W_2 = \frac{30 \times 1.020}{\sqrt{3}} = 17.66$$

$$\therefore, W_1 - W_2 = 17.66$$

$$\therefore, W_1 - 30 + W_1 = 17.66$$

$$\therefore, 2W_1 = 47.66$$

$$\therefore W_1 = 23.83 \text{ kW}$$

Also, $W_2 = 30 - 23.83 = 6.17 \text{ KW}$